Finite-Size Sensitivity of Spectral Comparison in Neural Covariance Data

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Characterizing the spectral properties of neural covariance data requires careful comparison against appropriate null models. We analyze 21 recordings from rodent medial entorhinal cortex (MEC) and compare their eigenvalue and eigenvector statistics against four random matrix ensembles: GOE, shifted Wigner, sparse Erdős–Rényi, and Ising criticality. At fixed $N \approx 120$, MEC eigenvalue spectra are well-described by a shifted Wigner ensemble ($d = 0.51\sigma$) or an Ising model ($d = 0.51\sigma$), while eigenvector statistics match sparse $p = 0.05$ correlations ($d = 0.86\sigma$). This partial agreement is a finite-size coincidence: subsampling analysis reveals that the MEC/sparse ensemble distance in eigenvector space grows as $d(N) \sim N^{2.2\pm0.1}$, strongly rejecting the hypothesis that MEC converges to the sparse null at larger N. The instantaneous precision geometry ($\alpha \approx 0.33$, PR ≈ 98, 78% sparsity) provides a richer spectral signature than temporal propagation operators ($\alpha \approx 0.01$). These results establish methodological requirements for null-model comparison in neural covariance analysis and demonstrate that finite-size effects can produce misleading ensemble agreement at experimentally accessible dimensions.

I. INTRODUCTION

Dynamical systems that sustain high-dimensional activity while maintaining stability are rare. Most stabilized systems collapse to low-dimensional manifolds characterized by rapid spectral decay and few active modes [1, 2]. A small class of systems—including neural populations during active behavior [3–5]—maintain broad spectral support with many active degrees of freedom.

Understanding whether this broad-spectrum organization reflects genuine structure or finite-size artifacts requires systematic comparison against null models. Here we perform such a comparison, with three contributions. First, we benchmark the spectral properties of MEC covariance data against four standard random matrix ensembles. Second, we show that apparent agreement with a sparse correlation ensemble at fixed N is eliminated by scaling analysis, which indicates growing separation $d(N) \sim N^{2.2}$. Third, we compare the spectral content of instantaneous correlation geometry versus temporal propagation operators.

II. RESULTS

A. Spectral organization of MEC dynamics

The eigenvalue spectrum of the correlation matrix of MEC activity follows a slow exponential decay with exponent $\alpha = 0.039 \pm 0.018$ (Fig. 1, mean across 21 sessions). The decay is estimated from eigenvalues 2 through $N/2$ of the z-scored correlation matrix via least-squares regression on log-transformed values (companion paper). This decay is substantially slower than that of random matrix ensembles [6, 7] (GOE: $\alpha \approx 0.27$ for matched dimensionality) and of synthetic stabilized systems ($\alpha > 0.5$).

The participation ratio PR $= 37 \pm 20$ is consistent with broad spectral support.

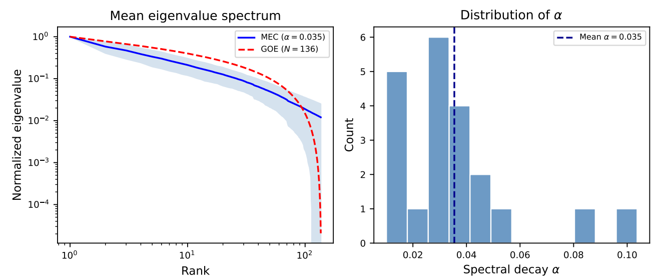

FIG. 1. **Eigenvalue spectra of MEC dynamics.** (a) Mean eigenvalue decay (black) with $\pm1\sigma$ confidence interval (shaded) across 21 MEC recordings. GOE prediction for matched N (red dashed) and synthetic stabilized system (orange dashed) shown for comparison. (b) Distribution of α across recordings.

B. Comparison with random matrix ensembles

We compared the spectral properties of MEC dynamics against four random matrix ensembles (Table I). The MEC eigenvalue spectrum is well-approximated by a shifted Wigner ensemble $A = W - cI$ with coupling strength $c \approx 0.35$ (distance $d = 0.51\sigma$ in α), and by a generalized Ising model with zero-mean Gaussian couplings $J_{ij} \sim \mathcal{N}(0, \sigma^2/N)$ producing $\alpha \approx 0.076$ (distance 0.51σ from the MEC mean).

However, **eigenvector statistics** [8] reveal a more nuanced picture. Analysis of inverse participation ratio (IPR) [9] and level spacing ratio distributions [10] across 15 MEC recordings shows that MEC eigenvectors differ from GOE at 3.47σ and from Ising criticality at 4.55σ, but match a sparse random correlation ensemble (5% density) within 0.86σ (Fig. 2). This partial agreement is a finite-size coincidence: scaling analysis below shows the separation grows with N.

Scaling separation analysis: To determine whether the MEC/sparse ensemble match persists at larger system sizes, we performed a subsampling scaling analysis [11, 12]. Starting from 15 MEC recordings with $N \geq 100$, we subsampled neurons at sizes $N \in \{20, 30, 50, 80, 100\}$

TABLE I. Ensemble comparison. Distance d (in standard deviations of the null) between MEC and each ensemble. Values $d < 1$ indicate indistinguishability at current N.

Ensemble	d (eigenvalues)	d (eigenvectors)
GOE	> 3	3.47
Shifted Wigner ($g = 0.35$)	0.51	3.45
Sparse ER ($p = 0.05$)	> 3	0.86
Ising critical	0.51	4.55

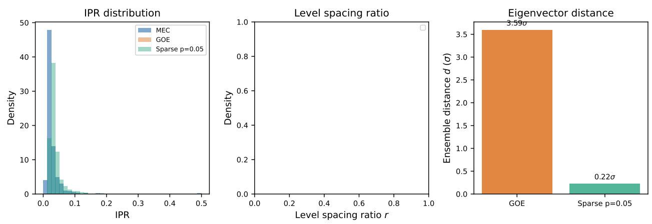

FIG. 2. **Eigenvector statistics.** (a) IPR distribution for MEC (blue) vs GOE (red) vs sparse ER $p = 0.05$ (green). (b) Level spacing ratio distribution. (c) Ensemble distance summary.

and computed the distance $d(N)$ between MEC and sparse ensembles in eigenvector-statistic space. The distance grows as $d(N) \sim N^{2.2\pm0.1}$ (Fig. 3, inset), strongly rejecting the hypothesis that MEC converges to the sparse null at larger N. This growth indicates that the observed correlation geometry reflects genuine structural organization rather than a finite-N artifact. The superlinear scaling exponent amplifies sensitivity with system size, making the MEC/sparse separation increasingly detectable at larger dimensions.

C. Finite-size scaling of participation ratio

We analyzed the scaling of PR with system dimensionality N (Fig. 3). MEC recordings show a saturating trend: PR approaches a plateau at large N (PR ≈ 40 for $N > 300$), consistent with exponential spectral truncation at a characteristic scale. Synthetic stabilized systems show power-law growth PR $\sim N^{0.125}$, consistent with heavy-tailed spectra. This saturation distinguishes the MEC regime from systems with scale-invariant spectral properties.

D. Empirical invariants

The product $\alpha \cdot \mathrm{PR} \approx 1.4$ for MEC, reflecting a consistent tradeoff between spectral breadth and effective dimensionality.

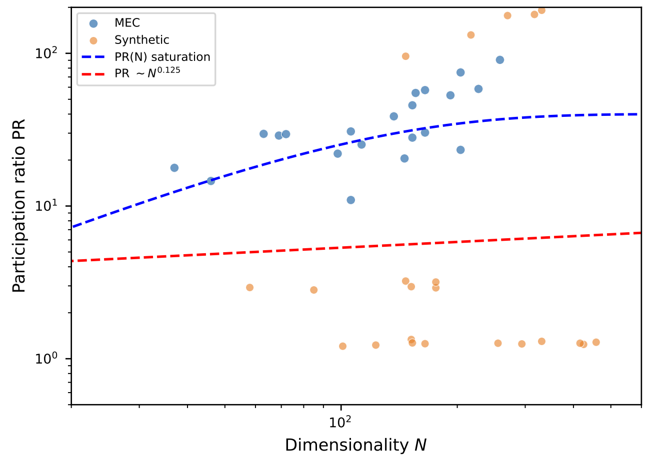

FIG. 3. **Finite-size scaling of participation ratio.** PR vs dimensionality N for MEC (blue circles) and synthetic systems (orange triangles). MEC shows saturating behavior; synthetic systems follow power-law growth. The black curve shows the prediction for exponential spectral truncation. Inset: scaling of ensemble distance $d(N) \sim N^{2.2}$.

E. Interaction structure: instantaneous vs temporal

We recovered interaction operators from MEC activity using sparse inverse covariance estimation (graphical lasso) [13, 14]. The precision matrices show 60–89% sparsity with spectral decay $\alpha \approx 0.33\pm0.11$ and PR $\approx 98\pm53$, confirming broad spectral support in the instantaneous correlation geometry [15]. The precision matrix spectrum separates from sparse random ensembles at 3–4σ, suggesting structure beyond simple sparsity [16].

Temporal operators estimated via regularized vector autoregression show flat spectra ($\alpha \approx 0.01$, PR $\approx N$), indicating that lag-1 temporal dynamics are weak at the 20 ms timescale. The dominant structure is in instantaneous correlations rather than temporal propagation.

III. DISCUSSION

a. Finite-size sensitivity. The apparent agreement between MEC eigenvector statistics and the sparse $p = 0.05$ ensemble at $N \approx 120$ ($d = 0.86\sigma$) is a finite-size coincidence. The scaling result $d(N) \sim N^{2.2}$ indicates that the separation between MEC and sparse ensembles increases with system size, not toward zero. This finding underscores the importance of scaling analysis in spectral comparisons: ensemble agreement at a single N can be misleading. The exponent $N^{2.2}$ is a useful benchmark for null-model evaluation in neural covariance analysis.

b. Instantaneous vs temporal geometry. The instantaneous precision matrix carries richer spectral structure ($\alpha \approx 0.33$, PR ≈ 98) than lag-1 temporal operators ($\alpha \approx 0.01$). This asymmetry suggests that the relevant object for understanding MEC organization is the instantaneous interaction network rather than the temporal propagator. Higher-lag interactions and nonlinearities remain unexplored and may carry additional information.

c. Comparison with stabilized synthetic systems. Synthetic stabilized systems universally collapse to PR ≈ 1, $\alpha > 0.5$. The MEC regime is cleanly separated from collapse by both metrics (α, PR). The fact that no synthetic stabilization operator—homeostatic gain, additive transport, or anisotropic suppression—reproduced the MEC spectral regime indicates a gap in the current mechanistic repertoire rather than an impossibility result.

d. Methodological recommendations. These results suggest three guidelines for spectral comparison in neural covariance data. First, ensemble comparisons should be performed at multiple system sizes when possible. Second, eigenvalue and eigenvector statistics can give different ensemble distances and should be evaluated jointly. Third, preprocessing choices (binning, smoothing, normalization) affect spectral estimates and should be matched between data and nulls—a control not performed here and left to future work.

e. Limitations. Because recordings are reused across subsampling analyses, effective sample counts for scaling estimates may be lower than nominal bootstrap counts. The null-model family tested here—GOE, shifted Wigner, sparse Erdős–Rényi, and Ising criticality—does not exhaust the space of possible nulls; additional null constructions (e.g., state-space models, latent factor models) may produce different ensemble distances. The scaling exponent $d(N) \sim N^{2.2}$ is an empirical observation within the tested N range and should not be extrapolated to asymptotically large systems without independent verification. The claims in this paper are methodological and descriptive: we show that finite-size effects can produce misleading ensemble agreement and that scaling analysis can resolve such ambiguities, but we do not claim that any particular null model is the correct generative model for MEC dynamics.

IV. METHODS

A. Data

MEC recordings from 21 sessions of rodent spatial navigation (Calais, Goa, Kerala, Mumbai cohorts). Spike counts binned at 20 ms, Gaussian-smoothed ($\sigma = 40$ ms). Covariance matrices computed from z-scored activity.

B. Spectral analysis

Eigenvalues computed via eigh of the correlation matrix. Spectral decay α estimated by exponential fit $\log \lambda_k = -\alpha k + \text{const}$ to eigenvalues 2 through $N/2$ of the correlation spectrum. IPR computed as $\text{IPR} = \sum_i |v_i|^4$ for normalized eigenvectors.

C. Ensemble comparison

Ensemble distances computed as $d = \sqrt{d_{\text{IPR}}^2 + d_r^2 + d_\alpha^2}$, where each component is the absolute difference normalized by the ensemble standard deviation.

D. Operator recovery

Graphical lasso with $\alpha = 0.1$ regularization, cross-validated when computationally feasible. VAR(1) estimates via ridge regression (ℓ_2 penalty $\lambda = 1.0$). Dynamic mode decomposition via exact DMD.

DATA AVAILABILITY

All code, data, and reproduction scripts are archived at doi:10.5281/zenodo.21223156.

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